

An exponential lower bound for the bit pigeonhole principle in resolution over parities

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With substantial assistance from Claude and GPT models.
More details in Section [10.3](#).

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Abstract

Resolution over parities, $\text{Res}(\oplus)$, extends resolution by allowing clauses whose literals are linear equations over $\mathbb{F}_2$. Superpolynomial size lower bounds were previously known only for restricted refutations: tree-like, regular, or of bounded depth. We prove that every DAG-like $\text{Res}(\oplus)$ refutation of the bit pigeonhole principle with $n + 1$ pigeons and $n = 2^\ell$ holes has more than $\exp(n/(32768 \ell^2)) = 2^{\Omega(n/\log^2 n)}$ clauses, for every $\ell \geq 32$, with no restriction on regularity or depth. The proof translates an arbitrary refutation with S clauses into a polynomial calculus refutation of degree $O(\log n)$ over $O(S + n^2)$ groups of extension variables in the style of Buss, Impagliazzo, Krajíček, Pudlák, Razborov, and Sgall. One substitution then removes all extension variables at once and leaves a nonzero low-degree polynomial derived from the pigeonhole axioms alone at degree $o(n)$; a degree lower bound in the style of Razborov, proved through the homology of chessboard complexes, shows that no such derivation exists. The argument also yields a general sufficient condition for $\text{Res}(\oplus)$ size lower bounds. The main theorem, this condition, and all their dependencies are formalized in Lean 4, and every statement links to its formal proof. The proof was developed with substantial AI assistance within an open research framework described in the final section.

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1 The claim and its setting

Resolution over parities augments ordinary resolution with affine equations over $\mathbb{F}_2$. The system was introduced by Itsykson and Sokolov [IS14, IS20]. Its unrestricted proof-size lower-bound problem remains an explicit open benchmark in the literature reviewed for this preprint [IPS26]. The contribution claimed here is an exponential lower bound for the standard bit pigeonhole principle in unrestricted DAG-like $\text{Res}(\oplus)$.

1.1 The bit pigeonhole principle

Fix $\ell \geq 2$, put $n = 2^\ell$ and $m = n + 1$, and introduce

$$b_{it} \quad (i \in [m], t \in [\ell]).$$

The row b_i encodes one of the n labels in $\{0, 1\}^\ell$. For a label z , let $[b_{it} \neq z_t]$ denote b_{it} when $z_t = 0$ and $\neg b_{it}$ when $z_t = 1$. The usual CNF encoding is

$$\text{BPHP}_n^{n+1} = \bigwedge_{\substack{i < i' \in [m] \\ z \in \{0,1\}^\ell}} \left(\bigvee_{t \in [\ell]} [b_{it} \neq z_t] \vee \bigvee_{t \in [\ell]} [b_{i't} \neq z_t] \right). \quad (1.1)$$

It has $v = m\ell$ variables and $n \binom{m}{2}$ clauses of width 2ℓ . No separate range or pigeon axiom is required: every Boolean row encodes exactly one available label.

1.2 The proof system

A *linear clause* is a disjunction of affine equations over $\mathbb{F}_2$. The basic rules are complementary-parity resolution and semantic weakening:

$$\frac{A \vee (u = 0) \quad B \vee (u = 1)}{A \vee B}, \quad \frac{C}{D} \quad \text{if } C \models D. \quad (1.2)$$

The pivot u may be any affine form. A refutation is a finite acyclic derivation from initial clauses to the empty clause. Its size is the number of nodes, including initial nodes, in its proof DAG. Earlier nodes may be used repeatedly, and no restriction is placed on proof depth or on repeated use of related pivots.

We also consider the convention permitting every sound inference from at most two linear-clause premises. Lemma 7.6 reduces this convention to (1.2) with constant node overhead. Fresh unconstrained propositional variables, if used, may be set to constants throughout a proof. Extension axioms defining new predicates are not part of the system considered here.

Theorem 1.1 (Main theorem). [Lean] *For every $\ell \geq 32$, with $n = 2^\ell$, every DAG-like $\text{Res}(\oplus)$ refutation of BPHP_n^{n+1} has more than*

$$\exp\left(\frac{n}{32768 \ell^2}\right)$$

nodes. The same bound holds with base two in place of e , and for both rule conventions above.

In terms of n the bound is $2^{\Omega(n/\log^2 n)}$. The constant $32768 = 2^{15}$ and the threshold 32 are what the formal proof verifies; no attempt was made to optimize them. Version 1 of this preprint stated only the following qualitative form, which was formalized first and which the main theorem implies for large ℓ .

Corollary 1.2 (Superpolynomial form). [Lean] *For every fixed real $K > 0$, there is $\ell_0(K)$ such that for every $\ell \geq \ell_0(K)$, every DAG-like $\text{Res}(\oplus)$ refutation of $\text{BPHP}_{2^\ell}^{2^\ell+1}$ has more than $2^{K\ell} = n^K$ nodes.*

The input length in (1.1) is polynomial in n , so the bound is also exponential in a fixed power of the input length, and it bounds the total proof-description size from below.

1.3 Prior results and the scope of the claim

Efremenko, Garlík, and Itsykson [EGI25] prove a $2^{\Omega(n^{1/3}/\log n)}$ size bound for regular $\text{Res}(\oplus)$ refutations of bit PHP. Their unrestricted DAG-like result is an affine-width bound. Bhattacharya, Chattopadhyay, and Dvořák [BCD24] exhibit formulas that are exponentially hard for bottom-regular $\text{Res}(\oplus)$, although they have short proofs in general ordinary resolution. This separation demonstrates the importance of removing regularity.

Bhattacharya and Chattopadhyay [BC25], Byramji and Impagliazzo [BI25], and the merged STOC 2026 paper [BCBI26] obtain exponential bounds in regimes allowing nearly quadratic proof depth. Their bounds retain a proof-depth restriction. Itsykson, Podolskii, and Shekhovtsov [IPS26] explicitly distinguish this progress from the general superpolynomial size problem; their unrestricted size consequence is quadratic.

Alekseev and Gaevoy [AG26] study constrained bit PHP. Their polynomial-depth bounds for general $\text{Res}(\oplus)$ are conditional; their unconditional results retain either a reversible-system restriction or a near-quadratic depth bound. They therefore do not give the unrestricted usual-bit-PHP size conclusion considered here.

Resolution over linear equations. $\text{Res}(\oplus)$ is the characteristic-two member of the family $\text{Res}(\text{lin}_R)$ of resolution over linear equations in a ring R , introduced by Raz and Tzameret [RT08]. Part and Tzameret [PT21] proved the first superpolynomial lower bounds for DAG-like $\text{Res}(\text{lin}_R)$: the subset-sum principle requires exponential size over the rationals. Their technique uses large characteristic and does not apply over $\mathbb{F}_2$. Khaniki [Kha22] proved almost quadratic lower bounds for DAG-like resolution over low-degree polynomial equations over finite fields, which includes resolution over linear equations, for mod- q Tseitin formulas and random k -CNFs. To our knowledge these were the first superlinear DAG-like bounds for such systems over finite fields; [EGI25] notes that the rules considered there differ from the formulation of $\text{Res}(\oplus)$ used here.

Other work on $\text{Res}(\oplus)$. Gryaznov, Pudlák, and Talebanfard [GPT22] introduced regular $\text{Res}(\oplus)$ and related it to read-once linear branching programs. Alekseev and Itsykson [AI25] lift resolution depth to regular and bounded-depth $\text{Res}(\oplus)$ size, and Efremenko and Itsykson [EI25, EI26] sharpen the closure technique of [EGI25] and obtain strong bounds for bounded-depth $\text{Res}(\oplus)$. All of these retain a regularity or depth restriction.

Polynomial calculus and the pigeonhole principle. The degree lower bound at the base of our argument is Razborov’s theorem [Raz98] that polynomial calculus refutations of the pigeonhole principle need degree $n/2+1$ over every field, which also describes the low-degree consequences of the pigeonhole axioms. Related degree bounds and methods are due to Impagliazzo, Pudlák, and Sgall [IPS99], Alekhnovich and Razborov [AR03], and Mikša and Nordström [MN15]. Section 4 reproves the part of Razborov’s theorem that we need, over $\mathbb{F}_2$, by a different route through chessboard

complexes; we do not claim this degree range as new (see the discussion before Theorem 4.2). The extension variables are those of Buss, Impagliazzo, Krajíček, Pudlák, Razborov, and Sgall [BIKPRS].

Scope. The theorem here removes both regularity and proof-depth restrictions. Targeted searches of the primary literature on 15 and 17 September 2026 found no preceding theorem with that scope; this is not an exhaustive certification of novelty. The theorem concerns $\text{Res}(\oplus)$ only. It does not address unary PHP in $\text{Res}(\oplus)$, and the $\text{AC}^0[p]$ -Frege lower-bound problem remains open; Section 9 says what does and does not follow.

1.4 Changes in revision 1

Version 1 (15 September 2026) stated Corollary 1.2. Revision 1 adds the exponential bound of Theorem 1.1 (Section 8) and the generic sufficient condition of Section 9, both formalized, a proof overview (Section 2), and a fuller account of related work. The development history formerly in the last section was condensed. No error in the mathematics of version 1 was found or corrected; the registry slot count of Section 7 was renamed from N to N_{reg} to avoid a clash with the number of unary columns.

1.5 Organization

Section 2 gives an overview of the proof, fixes terminology and parameters, and works through one simulated inference. Sections 3–7 prove the components in full, and Section 8 closes the parameters. Section 9 states the generic criterion; it can be skipped. Appendix A proves the homological input behind the matching extension. A companion online research notebook records the development of the argument and supporting experiments [Repo]; Section 10 describes these supplementary materials. The proofs presented here can be read independently of that notebook: none of its lemmas is required as an unexpanded black box in this preprint.

Every [Lean] link points to a source file in the published revision b47e9b1, rather than a moving branch. The linked files identify their exported statements and dependencies. The mathematical notation here is independent of Lean syntax.

2 Overview of the proof

This section describes the argument in words, fixes the terminology and the parameters, and works through the simulation of one inference. It contains no proofs; every claim made here is proved in the section cited for it.

Fix a $\text{Res}(\oplus)$ refutation of BPHP_n^{n+1} with S nodes. The proof has three steps.

2.1 Step 1: from a refutation to low-degree polynomial calculus

A linear clause C is a disjunction of affine equations. Write each equation through its *true-indicator* g_i , the affine form that is one exactly when the equation holds, so that C is falsified exactly on the affine flat $Z(C) = \{g_1 = \cdots = g_s = 0\}$. Following Buss, Impagliazzo, Krajíček, Pudlák, Razborov, and Sgall [BIKPRS], we attach to C fresh Boolean variables r_{ui} and the product

$$P_C = \prod_{u=1}^h \left(1 - \sum_{i=1}^s r_{ui} g_i \right),$$

together with the axioms $g_i P_C = 0$ and $r_{ui}^2 = r_{ui}$. On $Z(C)$ the product is one. At a point where C holds, each factor vanishes for half of the choices of its coefficients, so P_C behaves like the indicator of “ C is false” up to an error that halves with each of the h factors. The axioms $g_i P_C = 0$ say that P_C is supported on $Z(C)$. The line $P_C = 0$ is *not* an axiom: deriving it is how the simulation expresses that C has been proved.

What matters is that P_C has degree $2h$ whatever the width or the rank of C . The identity $1 - P_C = \sum_i U_{C,i} g_i$ (Corollary 7.2) lets a few local polynomial identities of degree at most $4h + 1$ simulate complementary-parity resolution and semantic weakening (Lemmas 7.3 and 7.4). Each simulated inference uses the *lines* $P_A = 0$, $P_B = 0$ derived earlier, never their derivations, so the degree does not grow along the proof. This is why neither the depth of the refutation nor regularity enters: the simulation is local and the proof DAG is only followed in topological order. An initial clause costs degree $2h + \ell$. Altogether (Theorem 7.9) the refutation becomes a polynomial calculus refutation of degree

$$D = \max\{2h + \ell, 4h + 1\}$$

from the compact pigeonhole axioms $\mathcal{Q}_{m,\ell}$ of (5.1) together with at most $N_{\text{reg}} = 3S + \binom{m}{2}$ blocks of extension axioms. With $h = 3\ell$ this is $D = 12\ell + 1 = O(\log n)$.

2.2 Step 2: removing all extension variables at once

A degree- $O(\log n)$ refutation with extension variables contradicts nothing by itself. We substitute polynomials of degree at most k in the original bit variables for all the r_{ui} , simultaneously, and replay the refutation (Lemma 3.4). A block whose inputs have rank $r \leq h(k + 1)$ can be substituted *exactly*: its h factors, each with coefficients of degree at most k , suffice to turn P_C into the true indicator of $Z(C)$, and the extension axioms become consequences of the Boolean axioms (Lemma 6.6). This is the origin of the rank threshold $h(k + 1)$.

A block of higher rank cannot be treated this way within degree k . Instead we look for a polynomial f of degree at most k whose restriction to the flat $Z(C)$ is identically zero. Then $f = \sum_i a_i g_i$ with $\deg a_i < k$, and substituting the a_i into the first factor and zero into the others turns P_C into $1 - f$. The extension axiom $g_i P_C$ becomes $g_i(1 - f)$, which is not a consequence of anything; but after multiplication by f it becomes $-g_i(f^2 - f)$, a consequence of the Boolean axioms. So we replay the whole refutation *multiplied by f* . Every axiom image is then derivable from $\mathcal{Q}_{m,\ell}$ alone, and the final line 1 becomes f (Lemma 6.7):

$$f \in \mathcal{C}_B(\mathcal{Q}_{m,\ell}), \quad B = k(D + 1).$$

This is why the end product is a derivation of a nonzero polynomial f and not of the constant one: the weight f has to vanish on the falsifying flats of all high-rank clauses at once, and the constant one does not.

A single f must serve every high-rank block. A flat of codimension $r > h(k + 1)$ has so few free coordinates that the restrictions of degree- k polynomials to it span a space smaller, by a factor $e^{-k^2/m}$, than the space $\mathcal{L}_k$ of *row-linear* polynomials (at most one bit variable from each pigeon). If

$$m \ln(4M) \leq k^2,$$

where M bounds the number of high-rank blocks, a dimension count gives a nonzero $f \in \mathcal{L}_k$ restricting to zero on all of them (Lemma 6.4). This inequality is the only place where the size of the refutation enters, through $M \leq N_{\text{reg}} = 3S + \binom{m}{2}$.

2.3 Step 3: a nonzero row-linear polynomial is not derivable

It remains to show that no nonzero $f \in \mathcal{L}_k$ has a polynomial calculus derivation of degree B from the pigeonhole axioms when B is below roughly $n/2$ (Theorem 5.7). This is stronger than the degree lower bound for refutations, which is the case $f = 1$. We decode bit labels into the unary (one variable per pigeon and hole) functional pigeonhole principle, where, as in Razborov’s analysis [Raz98], the linear functionals vanishing on all degree- B consequences are exactly the solutions of the matching-moment equations (4.1). Theorem 4.2 shows that every solution through a lower degree extends through degree B when $2B - 1$ is at most the number of holes; the extension problem is a cycle-filling problem in chessboard complexes, solved by the vanishing of their homology in the range of Björner, Lovász, Vrećica, and Živaljević [BLVZ94]. To separate a given f of degree t , a signed sum over a t -dimensional cube of partial assignments isolates one top-degree coefficient of f as the constant one, while mapping pigeonhole consequences to consequences of a smaller pigeonhole instance of degree $B - t$, where a normalized functional exists.

2.4 Parameters

The three steps meet in the inequalities of Theorem 6.8:

$$(n + 1) \ln(4M) \leq k^2, \quad 2k(D + 1) - 1 \leq n, \quad 4(k - 1) < n.$$

With $D = 12\ell + 1$, the second allows k up to about $n/(24\ell)$, and then the first holds as long as $\ln S$ is below a constant times n/ℓ^2 . This gives Theorem 1.1. The symbols used throughout are collected here.

ℓ , $n = 2^\ell$, $m = n + 1$, $v = m\ell$	bits per label, holes, pigeons (rows), bit variables
N	number of columns of a unary system ($N = n$ in the application)
S , $N_{\text{reg}} = 3S + \binom{m}{2}$	nodes of the refutation; blocks (slots) of the registry
h	number of factors of a block product (its <i>accuracy</i>); $h = 3\ell$
s	number of inputs of a block (clause width after compression, at most $v + 2$)
r	rank of a block: dimension of the span of its inputs; threshold $h(k + 1)$
D	degree of the simulating PC refutation; $D = 12\ell + 1$
k	degree bound for f and for the substituted coefficients
t	degree of a particular f ; number of rows in the separating cube; $t \leq k$
M	bound on the number of proper high-rank blocks; $M \leq N_{\text{reg}}$
$B = k(D + 1)$	degree at which f would be derived; needs $2B - 1 \leq n$

2.5 Terminology

Some terms come from the research notebook in which the proof was developed and are kept because the formal statements use them. *Old* variables are the bit variables b_{it} of the formula (in Section 4, the unary variables), as opposed to the fresh coefficient variables r_{ui} ; the *old base* is the axiom system before blocks are added, and an *old-base consequence* is a polynomial derivable from it alone. A *complete affine extension block of accuracy h* is the package (6.1): affine inputs g_i in the old variables, fresh coefficients, all *companions* $g_i P$, and all coefficient Boolean equations; “complete”

means that every companion is present, and “one-level” that no block’s inputs mention another block’s variables. A *registry* is a family of blocks fixed in advance, one *slot* per clause that the simulation may need. A polynomial is *row-linear* if each monomial has at most one bit variable from each pigeon. $\mathcal{I}_B$ and $\mathcal{C}_B$ denote degree- B Nullstellensatz and polynomial calculus consequences (Section 3); “NS” and “PC” abbreviate the two systems.

2.6 A worked example: one resolution step

Take accuracy $h = 1$ and resolve $A = (u = 1)$ with $B = (u = 0)$, for an affine form u , to the empty clause. The true-indicators are u for A and $1 - u$ for B , so with coefficients r and s

$$P_A = 1 - ru, \quad P_B = 1 - s(1 - u), \quad P_\emptyset = 1.$$

Suppose the lines P_A and P_B have been derived. Since $u^2 - u$ is a combination of Boolean axioms, polynomial calculus derives

$$(1 - u)P_A - r(u^2 - u) = (1 - u)(1 - ru) - r(u^2 - u) = 1 - u,$$

which is (7.4) in this case: from “ A holds” we have derived that the indicator of B is zero. Multiplying by s and adding P_B gives

$$s(1 - u) + (1 - s(1 - u)) = 1,$$

the value of the empty clause, and the refutation is complete. All lines have degree at most three. With side literals present, the same computation is carried out after multiplying by the product P_T of the conclusion T , and the side literals are absorbed by the companions of T ; with h factors, r and s are replaced by the polynomials $U_{A,u}$ and $U_{B,1-u}$ of degree $2h - 1$, which gives the bound $4h + 1$ of Lemma 7.3.

2.7 Why the restriction-based approaches stop short

As we understand the earlier lower bounds for regular and bounded-depth $\text{Res}(\oplus)$ [EGI25, BCD24, BC25, BI25, BCBI26, AI25, EI25], they follow a path through the refutation and maintain a set of linear forms (a closure, or the state of a game or of a random walk) that records what the path has learned. Regularity or a depth bound controls how much such a path can accumulate; a general DAG may reuse a node along many long paths, and the control is lost. The present argument never follows a path. It is static and algebraic: the whole DAG is translated into one polynomial calculus refutation whose degree does not depend on depth, in the manner of the extension-variable method of [BIKPRS]. That method was designed for bounded-depth Frege systems with counting gates, where it reduces lower bounds to degree bounds for systems with extension axioms; in that generality no low-degree way of removing the extension variables is known. For $\text{Res}(\oplus)$ the extension axioms have inputs that are affine in the original variables, and this is exactly what makes the one-shot removal of Step 2 possible: low rank allows exact substitution, and high rank leaves room for a common annihilating polynomial. The price is that the base system must exclude every nonzero row-linear consequence, not only the constant one, which is the Razborov-style statement of Step 3. For nested extension axioms, as they arise from $\text{AC}^0[p]$ -Frege proofs, we know of no corresponding removal step.

3 Ordinary degree and polynomial calculus

Throughout, the field is $K = \mathbb{F}_2$ and variable sets are finite.

Remark 3.1 (Generality). Several statements in Sections 3–7 were formalized in greater generality than is used here: over an arbitrary field, for infinite variable sets or infinite-dimensional spaces, or in arbitrary commutative rings. The linked Lean files record the exact hypotheses. One distinction made visible by formalization is used below: the degree of a block product or companion is in general only bounded above by the stated value, with equality under the hypotheses of Lemma 6.2. The argument only uses the upper bounds.

Degree bounds and accuracy parameters are integers. Polynomials are ordinary commutative polynomials. In particular, powers are not silently reduced using Booleanity. We use the nonnegative total-degree convention $\deg 0 = 0$. The Boolean equations are

$$\text{BOOL}_x = \{x_j^2 - x_j : j\}.$$

Every displayed generator is an equation with right side zero.

For a finite polynomial system $\mathcal{F}$, write

$$\mathcal{I}_B(\mathcal{F}) = \text{span}_K\{qF : F \in \mathcal{F}, \deg(qF) \leq B\} \subseteq \mathcal{P}_{\leq B}. \quad (3.1)$$

The degree constraint is imposed on each generator multiple. Cancellation between larger-degree multiples does not make them legal in $\mathcal{I}_B$.

Ordinary PC derives axioms of $\mathcal{F}$, linear combinations of previous lines, and $x_j g$ from a previous line g . A degree- B derivation has no line of degree exceeding B . Let $\mathcal{C}_B(\mathcal{F})$ be the space of polynomials derivable in this way. Refutation means deriving one. The space $\mathcal{C}_B$ is closed under linear combinations, but equality with $\mathcal{I}_B$ requires a proof.

Lemma 3.2 (Multiplying a completed line). [\[Lean\]](#) *If g has a degree- B PC derivation and q is any polynomial, then qg has a PC derivation through $\max\{B, \deg q + \deg g\}$. Moreover $\mathcal{I}_B(\mathcal{F}) \subseteq \mathcal{C}_B(\mathcal{F})$, and $q\mathcal{I}_B(\mathcal{F}) \subseteq \mathcal{I}_{B+\deg q}(\mathcal{F})$.*

Proof. Expand q into monomials. Starting from the already derived line g , multiply by the variables of each monomial one at a time, and then take the indicated linear combination. Each new line has degree at most $\deg g + \deg q$. The zero case is immediate. Applying this construction to each legal axiom multiple in (3.1), then summing, proves the second assertion. Multiplying each legal generator multiple by q proves the last assertion. There is no need to multiply the earlier derivation of g . $\square$

Lemma 3.3 (Degree-controlled Boolean reduction). [\[Lean\]](#) *For every polynomial R of degree at most d , its squarefree remainder $\overline{R}$ satisfies*

$$R - \overline{R} \in \mathcal{I}_d(\text{BOOL}_x).$$

If R vanishes at every Boolean point, then $R \in \mathcal{I}_d(\text{BOOL}_x)$. In particular, if $\deg q \leq k$, then

$$q^2 - q \in \mathcal{I}_{2k}(\text{BOOL}_x).$$

Proof. For $a \geq 2$,

$$x^a - x = (x^2 - x)(1 + x + \cdots + x^{a-2}).$$

Apply this identity to the powers in each monomial of R , one variable at a time. Every generator multiple has degree at most the original monomial degree, so the first assertion follows.

A squarefree polynomial that vanishes on $\{0, 1\}^v$ is zero. For example, write it as $A + x_v B$ with A, B independent of x_v . Evaluations at $x_v = 0, 1$, followed by induction on v , give $A = B = 0$. This proves the second assertion. Finally, every value of q at a Boolean point belongs to $\mathbb{F}_2$, where its square equals itself; apply the second assertion at degree $2k$. $\square$

The high-rank step of Section 6 needs literal membership in an affine ideal (Lemma 6.3), which pointwise vanishing does not supply.

Lemma 3.4 (Substitution and fixed-weight replay). [Lean] *Let ϕ substitute polynomials of degree at most $T \geq 0$ for source variables. Then $\deg \phi(p) \leq T \deg p$, and ϕ carries degree- D PC derivations to degree- TD derivations from the substituted axioms. More generally, fix a target polynomial w and a target system $\mathcal{G}$. If each source axiom F of degree at most D has its weighted image $w\phi(F)$ derivable from $\mathcal{G}$ through $TD + \deg w$, then so does the weighted image of every line of a degree- D source derivation.*

Proof. Expand into monomials to obtain the degree inequality. Induct on primitive PC steps. Zero, addition and scalar multiplication commute with the weighted map. At a nonzero multiplication $p \mapsto xp$, ordinary degree gives $\deg p + 1 \leq D$. The completed weighted line has degree at most $\deg w + T \deg p$; multiplying it by $\phi(x)$ costs at most $\deg w + T(\deg p + 1) \leq TD + \deg w$, by Lemma 3.2. A zero predecessor stays zero. Taking $w = 1$ and the substituted axioms as target system gives ordinary substitution. $\square$

Lemma 3.5 (Separation and compatible extension). [Lean: [extension](#) · [PC duality](#)] *Let U, S be subspaces of a vector space V over K . A linear functional on U extends to one on V vanishing on S if and only if it vanishes on $U \cap S$. In particular, if $p \notin S$, there is a linear functional λ with $\lambda(S) = 0$ and $\lambda(p) = 1$. Consequently, for either $S = \mathcal{I}_B(\mathcal{F})$ or $S = \mathcal{C}_B(\mathcal{F})$, a normalized annihilator on $\mathcal{P}_{\leq B}$ exists exactly when $1 \notin S$.*

Proof. Define the extension on $U + S$ by $\lambda(u + s) = \lambda(u)$. The intersection condition makes this well-defined. Extend a basis of $U + S$ to a basis of V , and assign zero on the added basis vectors. For separation, start on Kp with $\lambda(cp) = c$; its intersection with S is zero. Necessity follows by restricting a vanishing functional. The polynomial consequences follow by applying this construction inside $\mathcal{P}_{\leq B}$. $\square$

4 Matching moments and a stable old filtration

For $m \geq 0$ and $N \geq 1$, the functional unary system $\mathcal{F}_{m,N}^{\text{fun}}$ has variables X_{ij} and generators

$$\begin{aligned} & X_{ij}^2 - X_{ij}, \\ & X_{ij}X_{ij'} \quad (j \neq j'), \\ & X_{ij}X_{i'j} \quad (i \neq i'), \\ & \rho_i - 1, \quad \rho_i = \sum_{j=1}^N X_{ij}. \end{aligned}$$

The row exclusions in the second line are part of this auxiliary system. They are not inferred from a weaker unary encoding.

A matching T is a set of cells with distinct rows and distinct columns; write $X_T = \prod_{(i,j) \in T} X_{ij}$. The empty matching corresponds to the constant one.

Lemma 4.1 (Complete moment equations). [Lean: [normal form](#) · [moment equations](#)] *A linear functional on $\mathcal{P}_{\leq B}$ annihilating $\mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}})$ is exactly a choice of matching moments z_T , $|T| \leq B$, satisfying*

$$\sum_{j \notin \text{col}(T)} z_{T \cup \{(i,j)\}} = z_T \quad \text{for } |T| < B, \quad i \notin \text{row}(T). \quad (4.1)$$

Its value on an ordinary monomial is the moment of its squarefree support if that support is a matching, and is zero otherwise. Normalization is the additional condition $z_\emptyset = 1$.

Proof. Boolean equations reduce powers within the original degree. A squarefree monomial whose support has a repeated row or column contains one of the exclusion generators as a factor. Thus every polynomial reduces, within its degree, to a linear combination of matching monomials. Conversely, the stated rule for evaluating ordinary monomials annihilates every legal Boolean or exclusion multiple.

It remains to check row equations. In a row-equation multiple $q(\rho_i - 1)$ of degree at most B , we may first put q into the matching normal form: any error is a Boolean or exclusion consequence through $B - 1$, and multiplication by $\rho_i - 1$ keeps its certificate within B . For a matching multiplier X_T , if row i is already occupied, the product reduces to zero. Otherwise its reduction is

$$\sum_{j \notin \text{col}(T)} X_{T \cup \{(i,j)\}} - X_T.$$

These are precisely (4.1), and they exhaust the constraints. No positivity or multiplicativity is imposed on the functional. $\square$

Let $\Delta_{s,N}$ be the chessboard complex whose faces are matchings on an s -row, N -column board. The following consequence of the chessboard connectivity theorem of Björner, Lovász, Vrećica, and Živaljević [BLVZ94] is proved, with its homological prerequisite, in Appendix A:

$$s \geq 2, \quad N \geq 2s - 1 \quad \implies \quad \tilde{H}_{s-2}(\Delta_{s,N}; \mathbb{F}_2) = 0. \quad (4.2)$$

Status and idea of the next theorem. Theorem 4.2 and Corollary 4.3 say that, below degree about $N/2$, polynomial calculus over the functional pigeonhole axioms derives nothing beyond the Nullstellensatz span of the same degree, and that a consequence of low degree already has a certificate of its own degree. The degree range is that of Razborov’s lower bound [Raz98], whose proof also determines the low-degree consequences through an explicit basis; we do not claim the range, or the refutation degree bound it implies, as new. What we give is a different and self-contained proof over $\mathbb{F}_2$, in the form needed in Section 5: every functional given through degree k extends. We have not found this extension formulation, or its derivation from chessboard homology, in the literature, but we have not made an exhaustive search and claim only the proof as ours. The idea is that the unknown moments of degree s on a fixed set of s rows form an $(s - 1)$ -chain of the chessboard complex, the equations (4.1) prescribe its boundary, and the prescribed boundary is a cycle; the vanishing homology (4.2) supplies a filling.

Theorem 4.2 (Extension of prescribed matching moments). [Lean: [extension](#) · [annihilators](#)] Suppose $m \geq 0$, $N \geq \max\{1, 2B - 1\}$, and $0 \leq k \leq B$. Every functional annihilating $\mathcal{I}_k(\mathcal{F}_{m,N}^{\text{fun}})$ extends to one annihilating $\mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}})$. The constant moment is preserved. In particular normalized functionals exist through degree B .

Proof. Use Lemma 4.1. Starting from the prescribed moments, fill degrees $s = k + 1, \dots, B$. At degree one choose moments in each row whose sum is the prescribed constant moment; a column exists because $N \geq 1$.

For $s \geq 2$, fix a set S of s rows. Form the reduced $(s - 2)$ -chain

$$z_S = \sum_{\substack{T \text{ a matching on } S \times [N] \\ |T|=s-1}} z_T[T]$$

in $\Delta_{s,N}$. This is a cycle. At $s = 2$, its augmentation is the sum of two equal row totals, which is zero in $\mathbb{F}_2$. For $s > 2$, a face U of size $s - 2$ misses two rows of S . Its boundary coefficient is the sum of the prescribed marginal sums for those two rows, hence $z_U + z_U = 0$.

Since $N \geq 2s - 1$, equation (4.2) gives an $(s - 1)$ -chain y_S with boundary z_S . Assign the coefficients of y_S as the new moments on full s -matchings of $S \times [N]$. For a fixed $(s - 1)$ -matching, its boundary equation is exactly the missing-row equation in (4.1).

Different s -row sets have disjoint new unknowns: a size- s matching has a unique occupied row set. Their shared lower faces retain their already assigned values. Thus these fillings are globally consistent and complete the induction. If $s > m$, there are no new row sets or moments to fill. In particular, no hypothesis $m \geq B$ is required. The argument works for either value of the constant moment. Starting with $z_\emptyset = 1$ at degree zero supplies a normalized functional. $\square$

Corollary 4.3 (Old filtration stability and PC closure). [\[Lean\]](#) *Under $m \geq 0$ and $N \geq \max\{1, 2B - 1\}$,*

$$\mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}}) \cap \mathcal{P}_{\leq k} = \mathcal{I}_k(\mathcal{F}_{m,N}^{\text{fun}}) \quad (0 \leq k \leq B), \quad \mathcal{C}_B(\mathcal{F}_{m,N}^{\text{fun}}) = \mathcal{I}_B(\mathcal{F}_{m,N}^{\text{fun}}). \quad (4.3)$$

In particular $1 \notin \mathcal{C}_B(\mathcal{F}_{m,N}^{\text{fun}})$.

Proof. If a polynomial of degree at most k is outside $\mathcal{I}_k$, finite-dimensional linear duality gives a functional vanishing on $\mathcal{I}_k$ but not on that polynomial. Extend it by Theorem 4.2. The polynomial is then outside $\mathcal{I}_B$ as well. The reverse inclusion in the first equality is immediate.

To prove PC closure, induct over a degree- B derivation. Axioms and linear combinations lie in $\mathcal{I}_B$. At a nonzero multiplication step $g \mapsto X_{ij}g$, the predecessor has ordinary degree at most $B - 1$. If $g \in \mathcal{I}_B$, the first equality puts it in $\mathcal{I}_{B-1}$. Multiplication of its legal generator multiples by X_{ij} stays in $\mathcal{I}_B$. Hence $\mathcal{C}_B \subseteq \mathcal{I}_B$; the other direction is Lemma 3.2. A normalized annihilator excludes one. $\square$

5 The bit decoder and a separating functional

Here $n = 2^\ell$ with $\ell \geq 2$, and m is arbitrary. The compact bit system, our old base, is

$$\mathcal{Q}_{m,\ell} = \{E_{ii'} : i < i' \in [m]\} \cup \text{BOOL}_b, \quad E_{ii'} = \prod_{t=1}^{\ell} (1 - b_{it} - b_{i't}). \quad (5.1)$$

The degree of each collision generator is ℓ . Its vanishing asserts that the two bit rows encode different labels.

Identify unary columns with labels $z \in \{0, 1\}^\ell$, and define the ordinary ring map

$$\tau(b_{it}) = \sum_{z: z_t=1} X_{iz}. \quad (5.2)$$

Lemma 5.1 (Two-row interpolation). [\[Lean\]](#) *Let $\mathcal{U}_{i,i'}$ consist of Booleanity, same-row exclusions, and the two row-sum equations for distinct rows i, i' . For a polynomial R supported on these rows and of degree at most d ,*

$$R - \sum_{z,w} R(e_z, e_w) X_{iz} X_{i'w} \in \mathcal{I}_{\max\{d, 2\}}(\mathcal{U}_{i,i'}).$$

Here e_z, e_w are the corresponding one-hot assignments. Column exclusions are not used in this interpolation.

Proof. Reduce powers and same-row collisions. The remainder is a linear combination of 1, individual cells of the two rows, and products containing one cell from each row. This reduction costs at most d . Replace the constant by $\rho_i \rho_{i'}$, using

$$1 - \rho_i \rho_{i'} = (1 - \rho_i) + \rho_i(1 - \rho_{i'}),$$

and replace each remaining single cell by its product with the other row sum. These changes have certificates of degree at most two. The result is bilinear, and its coefficient at $X_{iz}X_{i'w}$ is its value at (e_z, e_w) , namely $R(e_z, e_w)$. $\square$

Lemma 5.2 (A polynomial left inverse for the decoder). [\[Lean\]](#) *For every $m, \ell \geq 0$, the decoder τ is injective and preserves ordinary total degree.*

Proof. Define a linear substitution λ on unary variables by sending X_{i,e_t} to b_{it} , where e_t is the unit bit label, and all other unary variables to zero. The unit labels are distinct and $\lambda\tau(b_{it}) = b_{it}$, so $\lambda\tau$ is the identity on polynomials. Both substitutions have degree at most one; applying the degree inequality in both directions proves equality, as well as injectivity. $\square$

Lemma 5.3 (Degree-preserving old-base transfer). [\[Lean\]](#) *For $m \geq 0$, $\ell \geq 2$, and every $B \geq 0$,*

$$q \in \mathcal{C}_B(\mathcal{Q}_{m,\ell}) \implies \tau q \in \mathcal{C}_B(\mathcal{F}_{m,n}^{\text{fun}}).$$

Proof. Bit Booleanity has a degree-two image certificate:

$$\tau(b_{it}^2 - b_{it}) = \sum_{z: z_t=1} (X_{iz}^2 - X_{iz}).$$

At a two-row one-hot assignment (e_z, e_w) , $\tau(E_{ii'})$ is one exactly when $z = w$. Lemma 5.1 therefore gives

$$\tau(E_{ii'}) - \sum_z X_{iz}X_{i'z} \in \mathcal{I}_\ell(\mathcal{U}_{i,i'}).$$

The remaining terms are column exclusions of degree two. Thus every axiom image has a certificate through its original degree. Replay the source proof using Lemma 3.4 with $T = 1$, $w = 1$. Only axioms admitted by the source degree ceiling are replayed, so $B \geq \ell$ need not be assumed. $\square$

Let $\mathcal{L}_k$ be the space spanned by monomials of degree at most k containing at most one bit variable from each row.

Lemma 5.4 (Row-linear dimension). [\[Lean\]](#) *For all $m, \ell, k \geq 0$,*

$$\dim \mathcal{L}_k = \sum_{j=0}^k \binom{m}{j} \ell^j. \tag{5.3}$$

Every nonzero polynomial in this space has a nonzero monomial coefficient of degree equal to its total degree.

Proof. Ordinary monomials form a basis. To choose a row-linear monomial of degree j , choose its j rows and one of ℓ coordinates in each. Terms with $j > m$ contribute zero. The last assertion follows by choosing a maximal-degree element of the finite nonempty support. $\square$

Lemma 5.5 (Binary cube degree drop). [Lean] Partition some source variables into t indexed groups. For each $\epsilon \in \mathbb{F}_2^t$, substitute scalars depending only on ϵ_i for variables in group i , and fixed affine target polynomials for all remaining variables. Write ϕ_ϵ for this substitution and $\Delta p = \sum_\epsilon \phi_\epsilon(p)$. Then

$$\deg \Delta p \leq \max\{\deg p - t, 0\}, \quad \deg p < t \implies \Delta p = 0.$$

If every source generator F has a cube-independent image r_F which is zero or a target generator, then

$$\Delta \mathcal{I}_B(\mathcal{F}) \subseteq \mathcal{I}_{\max\{B-t, 0\}}(\mathcal{G}).$$

Proof. A monomial missing one group has equal evaluations in pairs obtained by flipping that cube coordinate, so its sum is zero. A surviving monomial uses at least t selected variables, which become scalars; the remaining factors have total degree at most its original degree minus t . For a legal multiple qF , cube independence gives $\Delta(qF) = (\Delta q)r_F$. If $r_F = 0$ it vanishes; otherwise it is one target generator multiple, whose degree is at most $B - t$ when $B \geq t$. When $B < t$, it is zero. Extend by linearity. $\square$

Lemma 5.6 (Row-cube restriction and coefficient isolation). [Lean: [pair packing](#) · [restriction](#) · [coefficient isolation](#)] Choose t distinct rows and, for each, two labels differing only in one prescribed bit. If these pairs are disjoint, the one-hot substitutions on the selected rows, together with deletion of their $2t$ columns on other rows, give

$$\Delta \mathcal{I}_B(\mathcal{F}_{m,n}^{\text{fun}}) \subseteq \mathcal{I}_{\max\{B-t, 0\}}(\mathcal{F}_{m-t, n-2t}^{\text{fun}}).$$

For a row-linear f of degree at most t , this cube applied to τf is the constant coefficient of the monomial selecting those rows and prescribed bits. Such disjoint pairs can be chosen whenever $t \geq 1$ and $4(t-1) < 2^\ell$. More generally, the coefficient-isolation identity holds for arbitrary cube-independent polynomial images of the unselected bits, provided each selected bit has its prescribed base value plus its selected cube coordinate, and the other bits in that row stay constant.

Proof. On a selected row, each substitution is a one-hot assignment; on every other row, deleted columns become zero and surviving variables are relabeled. Booleanity, row exclusions and column exclusions consequently map to zero or the corresponding residual generator. Disjointness prevents a collision between selected rows or between selected and residual rows. A selected row-sum equation maps to zero, and every other row-sum equation maps to its residual counterpart. Lemma 5.5 applies.

A row-linear monomial missing a selected row cancels by a cube flip. If it uses all t selected rows, the degree bound forces it to use exactly one variable in each and none outside. The two-value sum in each row is one for the prescribed bit and zero for every other bit. This isolates exactly the claimed coefficient, independently of the images outside the selected rows. Finally, after selecting j pairs, their $2j$ vertices forbid at most $2j$ of the $2^{\ell-1}$ pairs in any specified direction. The strict packing bound permits the next choice. $\square$

Idea of the separation theorem. A degree lower bound says that the constant one is not derivable at degree B . Section 6 needs the same for every nonzero row-linear f of degree $t \leq k$, where B is much larger than k . Pick a monomial of top degree t in f ; it names t rows and one bit in each. For each of these rows choose two labels that differ exactly in the named bit, and sum over the 2^t ways of assigning one of its two labels to each selected row. Over $\mathbb{F}_2$ this signed sum Δ annihilates every monomial that misses a selected row, so it sends τf to the constant one (the

chosen coefficient), lowers every degree by t , and sends pigeonhole consequences to consequences of the pigeonhole principle on the remaining $m - t$ rows and $n - 2t$ columns. A normalized functional λ for that smaller instance, which exists by Theorem 4.2, then gives $\mu = \lambda \circ \Delta$.

Theorem 5.7 (Row-linear separation through a larger PC degree). [\[Lean\]](#) *Let $m \geq 0$, $\ell \geq 2$, $n = 2^\ell$, and suppose $1 \leq k \leq B$, $4(k - 1) < n$, and $2B - 1 \leq n$. For every nonzero $f \in \mathcal{L}_k$, there is a linear functional μ on unary polynomials through degree B such that*

$$\mu(\mathcal{C}_B(\mathcal{F}_{m,n}^{\text{fun}})) = 0, \quad \mu(\tau f) = 1.$$

It can be chosen with $\mu(1) = 0$ if $\deg f > 0$, and $\mu(1) = 1$ if $\deg f = 0$. Consequently

$$\tau f \notin \mathcal{C}_B(\mathcal{F}_{m,n}^{\text{fun}}), \quad f \notin \mathcal{C}_B(\mathcal{Q}_{m,\ell}).$$

Proof. A nonzero constant over $\mathbb{F}_2$ is one, so that case follows from a normalized matching functional. Otherwise put $t = \deg f$ and choose a nonzero degree- t coefficient. Lemma 5.6 supplies its coordinate pairs and residual operator Δ , with $\Delta \tau f = 1$. Because n is divisible by four, $4(k - 1) < n$ implies $k \leq n/4$. Thus $n - 2t \geq 1$, and

$$n - 2t \geq 2(B - t) - 1.$$

Theorem 4.2 supplies a normalized residual annihilator λ through $B - t$, even if there are fewer than $B - t$ remaining rows. Set $\mu = \lambda \circ \Delta$. To see that μ annihilates $\mathcal{C}_B(\mathcal{F}_{m,n}^{\text{fun}})$, note first that this space equals $\mathcal{I}_B(\mathcal{F}_{m,n}^{\text{fun}})$ by Corollary 4.3, which applies since $2B - 1 \leq n$. The first assertion of Lemma 5.6 gives $\Delta \mathcal{I}_B(\mathcal{F}_{m,n}^{\text{fun}}) \subseteq \mathcal{I}_{B-t}(\mathcal{F}_{m-t,n-2t}^{\text{fun}})$, and λ vanishes on the latter space. Coefficient isolation gives $\mu(\tau f) = 1$, whereas $\Delta 1 = 0$ gives $\mu(1) = 0$. Finally apply Lemma 5.3 for the bit consequence. $\square$

6 Excluding one complete affine extension level

The product construction belongs to the extension-Nullstellensatz approach of Buss, Impagliazzo, Krajíček, Pudlák, Razborov, and Sgall [\[BIKPRS\]](#). We use a complete one-level family and prove its required properties explicitly. An accuracy- h block has a finite tuple of old affine polynomials $g_1, \dots, g_s$, fresh coefficient variables r_{ui} , and product

$$P = \prod_{u=1}^h \left(1 - \sum_{i=1}^s r_{ui} g_i \right). \quad (6.1)$$

Its complete axioms are all companions $g_i P$ and all coefficient Boolean equations $r_{ui}^2 - r_{ui}$. The product P itself is not an axiom. Coefficients of different blocks are disjoint; every input depends only on the old variables. Rank means the dimension of the affine input span. A nonzero span is called proper if it does not contain one.

6.1 Affine linear algebra and ordinary restriction

Lemma 6.1 (Affine spans and coordinates). [\[Lean: linear algebra · affine spans · coordinates\]](#) *Let a finite affine system on $\mathbb{F}_2^v$ have span S . Its common zero set is nonempty exactly when $1 \notin S$. In that case it is an affine flat of codimension $r = \dim S$, every affine polynomial vanishing on it belongs to S , and any basis $F_1, \dots, F_r$ of S can be completed to invertible affine coordinates. Both coordinate substitutions preserve ordinary degree. If the zero set is empty, a constant linear combination of the inputs is one.*

Proof. Write the inputs as $g_i = c_i + L_i$. If a relation among their linear parts had nonzero constant part, their span would contain one. Otherwise all relations respect the right sides $-c_i$; linear algebra then solves $L_i(x) = -c_i$. Equivalently, define the consistent functional on the span of the L_i and extend it to the full dual space. When $1 \notin S$, independence of the F_j implies independence of their linear parts: a relation would be a constant in S , hence zero, and then all coefficients vanish. Complete those linear parts to a basis and retain the constants of the F_j . This gives an invertible affine coordinate change. The zero flat has first r coordinates zero, so an affine polynomial vanishing there is a linear combination of those coordinates. Substitution and its inverse have degree at most one, which proves degree preservation. Inconsistency is exactly the failed relation already identified, normalized to give one. $\square$

Lemma 6.2 (Scalar cleanup and exact block degrees). [Lean: [cleanup](#) · [exact degrees](#)] *Over $\mathbb{F}_2$, zero-span blocks, and unit-span blocks with $h \geq 1$, can be removed by constant coefficient substitutions without increasing PC degree. If a block has a genuine degree-one input, then $\deg P = 2h$; each nonzero degree-one input has companion degree $2h + 1$. In particular these degree equalities hold after discarding zero inputs in a proper nonzero block. Without these hypotheses they are only upper bounds.*

Proof. For zero span set all coefficients to zero. For unit span choose $\sum_i c_i g_i = 1$, use these constants in the first coefficient row, and use zero in the other rows. The product and companions become zero; over $\mathbb{F}_2$, every coefficient satisfies $c_i^2 = c_i$. Apply the substitution lemma with degree multiplier one to the remaining axioms. For the degree assertion, a genuine linear part contributes a nonzero degree-two term to each factor, since its coefficient variables are distinct. The polynomial ring is a domain, so degrees add in the product and on multiplication by a degree-one input. $\square$

Lemma 6.3 (Ordinary restriction dimension and ideal membership). [Lean: [dimension](#) · [ideal membership](#)] *An affine substitution into d variables maps a space of polynomials of degree at most k to a space of dimension at most $\binom{d+k}{k}$. Suppose the common zero flat of a finite affine system g_i is nonempty. If the ordinary polynomial restriction of f , of degree at most k , to that flat is zero, then for $k \geq 1$*

$$f = \sum_i a_i g_i, \quad \deg a_i \leq k - 1. \quad (6.2)$$

Here restriction means substitution into affine free coordinates, not merely evaluation at field points.

Proof. Substitution does not increase degree. The target monomials of degree at most k , counted by $\binom{d+k}{k}$, span the image. For ideal membership use the coordinates of Lemma 6.1. A polynomial whose restriction at $F_1 = \dots = F_r = 0$ is literally zero has each monomial divisible by at least one F_j . Assign each monomial to one such factor and remove it. The resulting cofactors have degree at most $k - 1$. Pull back the affine coordinate change and express each F_j as a constant combination of the original inputs. $\square$

Idea of the common kernel. The next lemma finds one row-linear f whose restriction to the zero flat of every proper high-rank block is the zero polynomial. It is a dimension count. Restriction to a flat of codimension r is a linear map from $\mathcal{L}_k$ into polynomials of degree at most k in $v - r$ free coordinates, a space of dimension $\binom{v-r+k}{k}$. For $r > 3\ell(k + 1)$ this is smaller than $\dim \mathcal{L}_k \geq \binom{m}{k} \ell^k$ by a factor $e^{-k^2/m}$. If M such maps are given and $Me^{-k^2/m} < 1$, their joint kernel is nonzero.

Lemma 6.4 (A common ordinary restriction kernel). [Lean: [parameter bounds](#) · [common kernel](#)] *Let $m > 0$, $\ell \geq 1$, $h = 3\ell$, $M, k \geq 1$, $k \leq m$, and*

$$m \ln(4M) \leq k^2.$$

In a finite family of affine blocks in $v = m\ell$ bits, suppose at most M blocks are proper and have rank greater than $h(k+1)$. There is a nonzero $f \in \mathcal{L}_k$ such that every block above that rank threshold either has one in its input span or admits (6.2). In each proper high-rank block, the ordinary restriction of f to its zero flat is zero.

Proof. Put $r_* = 3\ell(k+1) + 1$. If $r_* > v$, no proper high-rank block exists and choose $f = 1$. Otherwise its restriction uses at most $d = v - r_*$ free coordinates. The checked dimension comparison uses

$$k! \binom{d+k}{k} \leq (d+k)^k, \quad k! \binom{m}{k} \geq (m-k+1)^k.$$

Moreover $d+k \leq \ell(m-2k) \leq \ell(m-k+1)(1-k/m)$. The first inequality follows by inserting $d = m\ell - 3\ell(k+1) - 1$ and using $\ell \geq 1$; the second follows by expansion and $k \leq m$. Thus

$$\frac{\binom{d+k}{k}}{\binom{m}{k} \ell^k} \leq \left(1 - \frac{k}{m}\right)^k \leq e^{-k^2/m}.$$

Consequently the sum of the restriction-image dimensions is at most

$$M e^{-k^2/m} \binom{m}{k} \ell^k \leq \frac{1}{4} \binom{m}{k} \ell^k < \dim \mathcal{L}_k.$$

The joint restriction map has a nonzero kernel. Apply Lemma 6.3 to each proper high-rank block; unit-span blocks need no restriction condition. $\square$

Corollary 6.5 (A convenient kernel parameter). [Lean: [parameter bounds](#) · [common kernel](#)] *For $m, M \geq 1$, the choice $k = \lceil \sqrt{m \ln(4M)} \rceil$ is positive and satisfies the square condition of Lemma 6.4. Thus that lemma applies whenever additionally $k \leq m$.*

Proof. The radicand is positive; square the inequality defining the ceiling. $\square$

6.2 One simultaneous weighted substitution

The two substitutions of this subsection were described in Section 2: a low-rank block is replaced exactly by the indicator of its zero flat, a high-rank block by $1 - f$, and the whole refutation is replayed with weight f . The degree accounting below shows that every axiom image is derivable from the old base through $kD + \deg f$, after which Lemma 3.4 does the rest.

Lemma 6.6 (Low-rank packing without a retained core). [Lean] *For affine inputs over $\mathbb{F}_2$ of span rank $r \leq h(k+1)$, with $h, k \geq 0$, there are coefficient polynomials of degree at most k substituting the block product to*

$$Z = \prod_{j=1}^r (1 - F_j),$$

where F_j is a basis of the input span. This polynomial has degree at most r , is the Boolean indicator of the common zero set, and $g_i Z \in \mathcal{I}_{r+1}(\text{BOOL})$ for every input. No properness assumption is needed.

Proof. Partition the basis into h ordered bins of size at most $k + 1$. For each bin J , use

$$1 - \prod_{j \in J} (1 - F_j) = \sum_{j \in J} F_j \prod_{t \in J, t < j} (1 - F_t).$$

Express each F_j as a constant combination of the inputs. Every coefficient has at most k affine factors. Multiplying the resulting bin products gives Z . At every Boolean point it is one precisely when the whole input span vanishes, so $g_i Z$ vanishes everywhere and has degree at most $r + 1$. Apply Boolean reduction. If $h = 0$, the rank bound forces $r = 0$, all inputs are zero, and the empty product is one. $\square$

Lemma 6.7 (Weighted removal, including unit-span blocks). [\[Lean\]](#) *Let an old system $\mathcal{F}$ in finitely many variables over $\mathbb{F}_2$ contain all old Boolean equations. Adjoin complete accuracy- h blocks with finite affine input tuples and disjoint fresh coefficients, where $h, k \geq 1$. Suppose this system has a degree- D ordinary-PC refutation, $D \geq 2h + 1$. Let f be any ordinary polynomial of degree $s \leq k$. For every block of rank greater than $h(k + 1)$, suppose either its input span contains one or it admits (6.2). Then*

$$f \in \mathcal{C}_{k(D+1)}(\mathcal{F}).$$

No multilinearity or properness is required of f or of the blocks.

Proof. Make one simultaneous substitution fixing the old variables, with all coefficient images of degree at most k . For a low-rank block use Lemma 6.6. Its weighted companions have Boolean certificates through $s + r + 1 \leq s + h(k + 1) + 1 \leq s + kD$. For a high unit-span block, use the constant substitution making its product zero. For every other high block use the cofactors of (6.2) in the first row and zero elsewhere. Its product becomes $1 - f$, and a weighted companion becomes

$$f g_i (1 - f) = -g_i (f^2 - f).$$

Its Boolean certificate costs at most $2s + 1 \leq s + kD$. A coefficient image β has weighted Boolean equation $f(\beta^2 - \beta)$, certified through $s + 2k \leq s + kD$. An old axiom of degree $e \leq D$ has weighted image certified through $s + e \leq s + kD$. These estimates initialize every source axiom used by the refutation at the common ceiling $kD + s$, irrespective of any companion whose actual degree is below its general upper bound. Lemma 3.4, with $T = k$ and weight f , replays the whole primitive derivation. The final image of one is f . $\square$

Theorem 6.8 (Finite-parameter affine exclusion). [\[Lean\]](#) *Let $m > 0$, $\ell \geq 2$, $n = 2^\ell$, $h = 3\ell$, and let a finite complete old-affine family be adjoined to $\mathcal{Q}_{m,\ell}$. Suppose $M, k \geq 1$, at most M blocks are proper and of rank greater than $h(k + 1)$, and, with $B = k(D + 1)$,*

$$m \ln(4M) \leq k^2, \quad k \leq m, \quad D \geq 2h + 1, \quad 2B - 1 \leq n, \quad 4(k - 1) < n. \quad (6.3)$$

Then the augmented system has no ordinary-PC refutation through degree D .

Proof. The common-kernel lemma supplies nonzero $f \in \mathcal{L}_k$. Weighted removal would derive f from the old base through B , contrary to Theorem 5.7. Zero and unit spans are already covered by removal, so no preliminary change of the family is needed. $\square$

Lemma 6.9 (Uniform eventual parameter room). [\[Lean\]](#) *Fix natural numbers a, A, d . Put $n = 2^\ell$, $t = \ell + 1$, $q = \sqrt{2}$, and $k = \lceil tq^\ell \rceil$. For every sufficiently large ℓ , simultaneously*

$$\ell \geq 2, \quad 1 \leq k \leq n + 1, \quad (n + 1) \ln(4(A2^{a\ell} + 1)) \leq k^2,$$

$$4(k-1) < n, \quad 2k(A(\ell+1)^d + 1) - 1 \leq n.$$

The threshold depends only on a, A, d .

Proof. The ceiling gives $tq^\ell \leq k \leq 2tq^\ell$. Set $C = \ln(4(A+1))$. Since $\ln 2 \leq 1$, $\ln(4(A2^{a\ell}+1)) \leq C + a\ell$. For sufficiently large ℓ , $2(C + a\ell) \leq t^2$. Together with $n+1 \leq 2n$ and $q^{2\ell} = n$, this proves the square condition. For every fixed natural e , $t^e/q^\ell \rightarrow 0$: its successive ratio tends to $1/q < 1$, so its tail is dominated by a geometric sequence. Applying this to $e = d+1$ gives eventually

$$4k(At^d + 1) \leq 8(A+1)t^{d+1}q^\ell < n.$$

This implies the remaining degree and packing bounds, and also $k \leq n+1$. $\square$

Corollary 6.10 (Polynomial inventory and polylogarithmic degree). [\[Lean\]](#) Fix natural a, A, d . For all sufficiently large ℓ , every finite complete accuracy- 3ℓ affine family over $\mathcal{Q}_{2^{\ell+1}, \ell}$ with at most $A2^{a\ell} + 1$ proper blocks has no PC refutation of degree $D \leq A(\ell+1)^d$.

Proof. Raise the proposed degree ceiling to $D' = \max\{D, 6\ell+1\}$. It is at most $(A+7)(\ell+1)^{d+1}$. Apply Lemma 6.9 with constants $a, A+7, d+1$, and use its k and the upper inventory bound in Theorem 6.8. All required inequalities hold uniformly over the actual families. $\square$

7 A direct PC simulation of the proof DAG

Unless stated otherwise, $h \geq 1$ and there are finitely many old variables, say v . Represent a linear clause by its affine true-indicators:

$$C = \bigvee_i (g_{C,i} = 1), \quad Z(C) = \{b : g_{C,i}(b) = 0 \text{ for all } i\}.$$

An equation $a(b) = \alpha$ has true-indicator $1 + a(b) + \alpha$. The set $Z(C)$ is an affine subspace or empty.

Assign to each nonempty clause a fresh complete accuracy- h block, and write its product as P_C . Set $P_\emptyset = 1$, using no block for the empty clause. [\[Lean\]](#) Fix the whole block family before following the proof DAG. Clause values $P_C = 0$ will be derived; they are not included among its axioms.

Lemma 7.1 (General product telescoping). [\[Lean\]](#) For polynomials g_i, r_{ui} with finitely many indices i , and $h \geq 0$, put

$$P_v = \prod_{u=0}^{v-1} \left(1 - \sum_i r_{ui} g_i\right), \quad U_{h,i} = \sum_{v=0}^{h-1} r_{vi} P_v.$$

Then $1 - P_h = \sum_i U_{h,i} g_i$. If $\deg g_i \leq \delta$, $\deg r_{vi} \leq 1$, and $h \geq 1$, then

$$\deg P_h \leq h(\delta+1), \quad \deg U_{h,i} \leq 1 + (h-1)(\delta+1), \quad \deg(g_i P_h) \leq \deg g_i + h(\delta+1).$$

These are upper bounds; no freshness or nonvanishing of the inputs is assumed.

Proof. The identity holds at $h=0$. Writing $s_h = \sum_i r_{hi} g_i$, the recurrences $P_{h+1} = P_h(1 - s_h)$ and $U_{h+1,i} = U_{h,i} + r_{hi} P_h$ give

$$1 - P_{h+1} = 1 - P_h + P_h s_h = \sum_i U_{h+1,i} g_i.$$

Each factor has degree at most $\delta+1$, and each summand of $U_{h,i}$ has degree at most $1 + (h-1)(\delta+1)$. The sum and product degree inequalities prove the three bounds. At $h=0$, $P_0 = 1$ and $U_{0,i} = 0$. $\square$

Corollary 7.2 (Affine-clause prefix identity). [\[Lean\]](#) Every accuracy- h clause block, $h \geq 1$, has coefficients satisfying

$$1 - P_C = \sum_i U_{C,i} g_{C,i}, \quad \deg P_C \leq 2h, \quad \deg U_{C,i} \leq 2h - 1. \quad (7.1)$$

Its companions have degree at most $2h + 1$.

Proof. Apply Lemma 7.1 with $\delta = 1$, reindexing coefficient rows from zero-based to one-based. Exact companion degrees require the additional hypotheses in Lemma 6.2; a loose input-degree bound alone would not justify equality. $\square$

Lemma 7.3 (Complementary-parity resolution). [\[Lean\]](#) Suppose $A = (\bigvee_i (a_i = 1)) \vee (u = 1)$ and $B = (\bigvee_j (b_j = 1)) \vee (1 - u = 1)$, with u affine. Let T have the union of the context inputs a_i, b_j . If P_A, P_B have PC derivations through degree K , then P_T has a derivation in the same fixed block system through $\max\{K, 4h + 1\}$.

Proof. The context companions $a_i P_T, b_j P_T$ are axioms of the conclusion block; a repeated input may use the same axiom. The prefix identities yield

$$R_A = P_T - U_{A,u} u P_T = P_A P_T + \sum_i U_{A,i} (a_i P_T), \quad (7.2)$$

$$R_B = P_T - U_{B,1-u} (1 - u) P_T = P_B P_T + \sum_j U_{B,j} (b_j P_T). \quad (7.3)$$

Lemma 3.2 derives these through $\max\{K, 4h\}$. In particular it multiplies the completed lines P_A, P_B , not their entire earlier derivations.

For an affine $u = c + \sum_t c_t b_t$, the identity

$$u^2 - u = \sum_t c_t (b_t^2 - b_t)$$

has a degree-two proof. Therefore

$$(1 - u) R_A - U_{A,u} P_T (u^2 - u) = (1 - u) P_T \quad (7.4)$$

is derivable through $\max\{K, 4h + 1\}$. The right side is a completed line of degree at most $2h + 1$. Multiply it by $U_{B,1-u}$, costing at most $4h$, and add (7.3). The result is P_T . All multiplications expand into primitive PC steps by Lemma 3.2. $\square$

Lemma 7.4 (Semantic weakening and tautologies). [\[Lean\]](#) If $C \models D$ and P_C has a degree- K derivation, then P_D is derivable through $\max\{K, 4h\}$. A tautological D has a direct value derivation through $2h + 1$.

Proof. If $Z(D)$ is nonempty, then $Z(D) \subseteq Z(C)$. Each $g_{C,i}$ vanishes on $Z(D)$, so affine linear algebra gives constants λ_{ij} with

$$g_{C,i} = \sum_j \lambda_{ij} g_{D,j}.$$

To see this directly, choose affine coordinates beginning with a basis of the equations defining $Z(D)$. An affine polynomial vanishing when these coordinates are zero has only those coordinate terms. Consequently $g_{C,i} P_D$ is a constant linear combination of conclusion companions. Now use

$$P_D = P_C P_D + \sum_i U_{C,i} (g_{C,i} P_D)$$

and Lemma 3.2, with ceiling $\max\{K, 4h\}$.

If $Z(D)$ is empty, its affine equations are inconsistent, and their span contains one. The same constant combination of their companions gives P_D , through $2h + 1$. If the conclusion is empty, its value is one; the preceding formulas still apply with no conclusion block. In the nonempty-zero-set case, an implication to the empty clause forces all premise indicators to be identically zero, so the already derived premise value is one. $\square$

Lemma 7.5 (Binary affine separator). [\[Lean\]](#) *Let V be a vector space over $\mathbb{F}_2$, and A, B, D finite clauses of affine true-indicators. Suppose $A \wedge B \models D$, but neither $A \models D$ nor $B \models D$. There is an indicator u already in A , taking both values on $W = Z(D)$, such that*

$$W \cap Z(A) = \{x \in W : u(x) = 0\}, \quad W \cap Z(B) = \{x \in W : u(x) = 1\}.$$

Proof. Choose $a \in W$ satisfying A and $b \in W$ satisfying B . Soundness forces $a \in Z(B)$ and $b \in Z(A)$. Choose indicators $u \in A$, $v \in B$ with $u(a) = v(b) = 1$; then $u(b) = v(a) = 0$. Soundness also gives $W \subseteq Z(A) \cup Z(B)$. If $x \in W \cap Z(B)$ had $u(x) = 0$, the point $a - b + x$ would belong to W and satisfy both $u = v = 1$, a contradiction. Here we used only the affine identity $g(a - b + x) = g(a) - g(b) + g(x)$. Thus $u = 1$ on $W \cap Z(B)$, while $u = 0$ on $W \cap Z(A)$. The covering inclusion gives the two reverse implications and hence the fiber equalities. The witnesses a, b give both values. $\square$

Lemma 7.6 (Two-premise semantic inference). [\[Lean\]](#) *Every sound inference $A \wedge B \models D$ between finite affine clauses admits a derivation using at most three semantic weakening or complementary-parity resolution steps, introducing at most two auxiliary clauses. It is either a one-premise weakening or has the form*

$$A \models D \vee (u = 1), \quad B \models D \vee (u = 0), \quad \frac{D \vee (u = 1) \quad D \vee (u = 0)}{D},$$

where u can be chosen from A . Semantic implication is preserved under any affine restriction of the underlying space.

Proof. If either premise implies D , use weakening. Otherwise apply Lemma 7.5: on $Z(D)$, satisfaction of A forces $u = 1$, and satisfaction of B forces $u = 0$. This proves the two weakenings; resolution concludes. Restriction simply precomposes the clauses with an affine map and preserves every implication. $\square$

Lemma 7.7 (Clause compression). [\[Lean\]](#) *On a finite-dimensional $\mathbb{F}_2$ -vector space of dimension v , every finite affine clause has a semantically equivalent subclause of width at most $v + 1$.*

Proof. The space of affine forms has dimension $v + 1$. Extract a basis of the span from the clause's own indicators. Simultaneous vanishing of the basis is equivalent to vanishing of the entire span, hence gives the same falsifying set. This works for tautologies as well as consistent systems and introduces no new indicators. $\square$

Idea of the registry theorem. The lemmas above simulate one inference at a time, given blocks for its premises and its conclusion. A polynomial calculus refutation needs all of its axioms fixed in advance, so the next theorem first reserves a block for every clause that can occur: one for each node, two for the auxiliary clauses that Lemma 7.6 may introduce at that node, and one for each supplied initial clause. It then walks through the DAG in topological order. Because each step uses only the already derived lines P_A , P_B and costs at most $4h + 1$, the degree of the whole derivation is the maximum, not the sum, of the local costs.

Theorem 7.8 (A fixed registry for a finite proof DAG). [\[Lean\]](#) *Consider an S -node affine-clause DAG on v old bits, using semantic weakening, complementary resolution, or sound binary inference. Suppose there is a finite supplied family $(A_j)_{j \in J}$, of widths at most w , such that each permitted initial clause is implied by some A_j . There is one complete accuracy- h registry with*

$$N_{\text{reg}} = 3S + |J| \text{ slots}, \quad W = \max\{w, v + 2\},$$

at most $N_{\text{reg}}W$ inputs and at most $v + hN_{\text{reg}}W$ variables. If the supplied initial values have PC derivations in this registry through $D \geq 4h + 1$, then every compressed source value has a derivation through that same D . An empty final clause yields a refutation.

Proof. Compress each source clause to width at most $v + 1$, preserving its semantics, and choose the plans from Lemma 7.6. Reserve one primary and two auxiliary slots per source node, plus one slot per supplied initial. An auxiliary clause adds one pivot indicator to a primary clause, so has width at most $v + 2$. Unused slots can contain the empty clause and contribute no inputs or coefficients. Each input has h distinct coefficient variables, giving the inventory. Fix these slots and all their complete axioms first. In a topological order, simulate each planned weakening or resolution with Lemmas 7.3 and 7.4. Initial coverage is another weakening. Reuse completed premise values; the degree ceiling never acquires a factor depending on proof height. $\square$

Fresh unconstrained propositional variables may be fixed to zero before this construction. Semantic implication survives restriction; a restricted constant pivot is a semantic weakening. This does not authorize extension axioms defining additional predicates.

Theorem 7.9 (Complete clause-to-PC transfer). [\[Lean\]](#) *For arbitrary $m, \ell \geq 0$, a refutation of the usual bit-PHP CNF on m rows and $n = 2^\ell$ labels with S nodes yields, for every integer $h \geq 1$, a complete one-level old-affine family over $\mathcal{Q}_{m,\ell}$ with at most $3S + \binom{m}{2}$ blocks and an ordinary-PC refutation through*

$$D = \max\{2h + \ell, 4h + 1\}. \quad (7.5)$$

With $v = m\ell$, $N_{\text{reg}} = 3S + \binom{m}{2}$, and $W = \max\{\ell, v + 2\}$, the total input count is at most $N_{\text{reg}}W$, and the total variable count is at most $v + hN_{\text{reg}}W$. There is no proof-height restriction.

Proof. [\[Lean\]](#) Introduce the compact initial linear clauses

$$C_{ii'} = \bigvee_{t=1}^{\ell} (b_{it} + b_{i't} = 1).$$

Each $C_{ii'}$ semantically implies each label-specific initial clause for the same pair in (1.1): both rows encoding the same label falsifies $C_{ii'}$. Use these $\binom{m}{2}$ clauses as the supplied family in Theorem 7.8; their widths are at most ℓ .

For a compact initial clause set $g_t = b_{it} + b_{i't}$ and $T_t = \prod_{s < t} (1 - g_s)$. Its value has the ordinary identity

$$P_{C_{ii'}} = E_{ii'} P_{C_{ii'}} + \sum_{t=1}^{\ell} T_t (g_t P_{C_{ii'}}). \quad (7.6)$$

The first term uses the old degree- ℓ collision axiom; the other terms use the block's companions. Every multiple has degree at most $2h + \ell$. This is an NS, hence PC, derivation at that degree.

Apply Theorem 7.8 in the source proof's acyclic order. Initial values are obtained from (7.6) and Lemma 7.4. Each remaining inference is simulated by Lemmas 7.3 and 7.4. Their premise values are

actual earlier derived lines, so the same global ceiling (7.5) persists along the DAG. The final empty clause has value one. All blocks have inputs affine in the old v bits and disjoint fresh coefficients. Their full companion and coefficient-domain families are present. No balancing or multiplication of an entire preceding derivation is used. $\square$

8 Closing the proof-size parameters

Both results of Section 1 combine Theorem 7.9 at $h = 3\ell$ with Theorem 6.8; they differ only in the choice of k .

Proof of Theorem 1.1. Let $\ell \geq 32$, $n = 2^\ell$, $m = n + 1$, and put $x = n/(32768\ell^2)$. Suppose a refutation has $S \leq e^x$ nodes. Theorem 7.9 with $h = 3\ell$ gives a complete affine family over $\mathcal{Q}_{m,\ell}$ with at most $M = 3S + \binom{n+1}{2}$ blocks and a PC refutation of degree $D = 12\ell + 1 \geq 2h + 1$. We verify (6.3) for

$$k = \left\lfloor \frac{n}{32\ell} \right\rfloor.$$

An induction on $\ell \geq 32$ shows $32768(2\ell + 4)\ell^2 \leq 2^\ell$, that is, $x \geq 2\ell + 4$. Hence $e^x \geq 2^{2\ell+4} = 16n^2$. Since $\binom{n+1}{2} \leq 4n^2$,

$$4M \leq 12e^x + 16n^2 \leq 13e^x \leq e^{2x}, \quad \ln(4M) \leq 2x.$$

From $n \geq 64\ell$ we get $k \geq 1$ and $k + 1 \leq 2k$, so $n < (k + 1)32\ell \leq 64\ell k$. Therefore

$$(n + 1)\ln(4M) \leq 2n \cdot 2x = \frac{n^2}{8192\ell^2} \leq \frac{n^2}{4096\ell^2} \leq k^2.$$

Moreover $32\ell k \leq n$ gives $k \leq m$, $4(k - 1) < n$, and

$$2B = 2k(12\ell + 2) \leq 32\ell k \leq n.$$

Theorem 6.8 now says that the family has no PC refutation of degree D , a contradiction. Hence $S > e^x \geq 2^x$. Both rule conventions are covered by Theorem 7.9. $\square$

Proof of Corollary 1.2. The corollary follows from Theorem 1.1, since $n/(32768\ell^2) \geq K\ell$ for large ℓ . We also record the independent argument of version 1, which is the one formalized in the linked file and uses the smaller k of Lemma 6.9. Fix $K > 0$ and let $a = \max\{\lceil K \rceil, 2\}$. Use Corollary 6.10 with parameters $a, 13, 1$, and take ℓ above its uniform threshold and at least two. Put $n = 2^\ell$. Suppose a refutation has $S \leq n^K$ nodes. Since $n \geq 1$ and $a \geq K$, $S \leq n^a$. Theorem 7.9, with $h = 3\ell$, constructs an actual complete affine registry with

$$N_{\text{reg}} \leq 3S + \binom{n+1}{2}, \quad D = 12\ell + 1.$$

Its initial clauses are those of the usual bit CNF, and its empty final clause gives a PC derivation of one from exactly the complete family over $\mathcal{Q}_{n+1,\ell}$. No additional initial-value hypothesis remains. Now

$$\binom{n+1}{2} \leq (n+1)^2 \leq 4n^2 \leq 4n^a, \quad N_{\text{reg}} \leq 7n^a \leq 13 \cdot 2^{a\ell} + 1, \quad D \leq 13(\ell + 1).$$

The number of proper blocks is at most N_{reg} , so these are precisely the inventory and degree bounds excluded by the corollary, a contradiction. The threshold is independent of the alleged proof. Therefore $S > n^K$.

This argument uses $k = \lceil (\ell + 1)(\sqrt{2})^\ell \rceil$, giving

$$k = O(\sqrt{n} \log n), \quad k(D + 1) = O(\sqrt{n}(\log n)^2) = o(n). \quad (8.1)$$

far below the room $k \approx n/(32\ell)$ used for Theorem 1.1. $\square$

8.1 What the conclusion uses

The source clauses may have any affine rank. The argument partitions the actual complete block family by rank, uses one common multiplier on the high side, and packs every block on the low side. The final contradiction is the impossibility of an old consequence $f \neq 0$ in a separated row-linear space. It is not an application of a degree lower bound for one alone.

Only the old functional system uses PC/NS equality, established by prescribed-moment extension. The augmented system is handled by a primitive PC replay. Similarly, only the number of source blocks and their compact input descriptions need be polynomial. Expanded PC polynomials and their derivations need not have polynomial monomial count: the contradiction uses degree.

8.2 Relationship to the broader PHP project

This result arose within a broader mathematical research project led by Kamil Braun, whose main objective is a superpolynomial lower bound for ordinary PHP in every fixed-depth $\text{AC}^0[p]$ -Frege system. That objective remains open. Theorem 1.1 concerns a specific subsystem over $\mathbb{F}_2$. The project's research record, supporting code, and autonomous-assistant workflow are maintained in *Noemesis*, an open framework for autonomous mathematical research and formal verification. Its source and research archive are publicly available in the `math-research` repository on GitHub [Repo]:

<https://github.com/kbr-/math-research>

Section 10 describes the archive and framework.

9 A generic sufficient condition

This section can be skipped. It records that Steps 1 and 2 of the proof never use the pigeonhole principle, and states what they give for an arbitrary set of linear clauses. Everything is over $\mathbb{F}_2$. Proofs are by reference to the bit-PHP proofs they abstract and to the formal proofs. The criterion was first extracted from the argument in an AI review of version 1 (Claude Fable 5.1, 16 September 2026, recorded in the research notebook) and was then formalized.

For a linear clause A with true-indicators $g_1, \dots, g_s$, its *clause polynomial* is $\prod_i (1 - g_i)$, of degree at most the width s ; it vanishes exactly at the assignments satisfying A . For a polynomial system $\mathcal{F}$, call a linear space U of polynomials *separated at degree B* if $U \cap \mathcal{C}_B(\mathcal{F}) = \{0\}$.

Theorem 9.1 (Generic criterion). [Lean: `PC level` · `proof size`] *Let $\mathcal{I}$ be a set of linear clauses in v variables, and let $(A_j)_{j \in J}$ be a finite family of linear clauses of width at most w such that*

every clause of $\mathcal{I}$ is semantically implied by some A_j . Let $\mathcal{F}$ be a polynomial system containing the Boolean equations and the clause polynomials of all A_j . Let $h, k \geq 1$ with $h(k+1) + 1 \leq v$, put

$$D = \max\{2h + w, 4h + 1\}, \quad B = k(D + 1),$$

and let U be a space of polynomials of degree at most k that is separated at degree B . Then every $\text{Res}(\oplus)$ refutation of $\mathcal{I}$ with S nodes, under either rule convention of Section 1, satisfies

$$\dim U \leq (3S + |J|) \binom{v - h(k+1) - 1 + k}{k}. \quad (9.1)$$

Proof by reference. Theorem 7.8 is already stated for arbitrary clause sets with a supplied covering family; it gives a registry with $3S + |J|$ slots. The initial values are derived as in (7.6), with the clause polynomial of A_j in place of $E_{ii'}$, through degree $2h + w$. This gives a PC refutation of degree D from $\mathcal{F}$ and the blocks. Lemma 6.7 is already stated for an arbitrary old system. In place of Lemma 6.4, restrict U to the zero flat of each proper block of rank above $h(k+1)$: by Lemma 6.3 each image has dimension at most $\binom{v - h(k+1) - 1 + k}{k}$, so if (9.1) failed, some nonzero $f \in U$ would restrict to zero on all of them, and Lemma 6.7 would give $f \in \mathcal{C}_B(\mathcal{F})$. $\square$

Corollary 9.2 (Density form). [Lean] Under the hypotheses of Theorem 9.1,

$$3S + |J| \geq \delta \exp\left(\frac{hk^2}{v+k}\right), \quad \delta = \frac{\dim U}{\binom{v+k}{k}}.$$

Proof. Removing r of v variables shrinks $\binom{v+k}{k}$ by at least the factor $\exp(-rk/(v+k))$, since each of the k factors $(v-r+j)/(v+j)$ is at most $1 - r/(v+k)$. Take $r = h(k+1) + 1 \geq hk$. $\square$

Here δ is the density of U among all polynomials of degree at most k . The bound is useful only if δ is not much smaller than $\exp(-hk^2/(v+k))$ while U remains separated at degree $k(D+1)$. For bit PHP, $U = \mathcal{L}_k$, $w = \ell$, $h = 3\ell$, the old system is $\mathcal{Q}_{m,\ell}$, the covering family is the $\binom{m}{2}$ compact clauses of Theorem 7.9, and separation is Theorem 5.7; the sharper count of Lemma 6.4 then gives Theorem 1.1. Thus Sections 6 and 7 are generic for any clause set whose covering clause polynomials are axioms of the base. What is specific to bit PHP is the separation theorem over the compact base, and the fact that a large separated space exists at all.

Remark 9.3 (A sufficient condition, not a tradeoff). Theorem 9.1 is not a relation between $\text{Res}(\oplus)$ size and PC refutation degree, nor between size and width. Its hypothesis is not that $\mathcal{F}$ has no low-degree refutation but that a large space U of low-degree polynomials meets the low-degree consequences only in zero. The conclusion of Step 2 is a nonzero derivable member of U , not a derivation of one, and nonzero polynomials can be legitimate low-degree consequences: the satisfiable system $\{x, x^2 - x\}$ derives x and does not derive one. We do not know a formula other than bit PHP for which a separated space of useful density has been established.

Remark 9.4 (Formulas with short refutations). [Lean] Read contrapositively, a clause set with a size- S refutation admits no separated space of degree k and dimension above the right side of (9.1). As a consistency check on the easiest parity contradiction: if $\mathcal{F}$ contains a polynomial L of degree at most one and also $1 - L$, then it derives one at degree one, every polynomial of degree at most B is derivable at degree $B \geq 1$, and the only separated space is zero. This checks one trivial family; it is not an audit of the known short $\text{Res}(\oplus)$ refutations.

9.1 What does not follow

Unary PHP. The standard substitution from unary PHP into bit PHP replaces a unary variable by the conjunction describing one bit label; see, for example, [BB21]. For bounded-depth Frege it has polynomial size overhead and a constant depth increase. But an affine form in unary variables becomes a polynomial in bit conjunctions, which is generally nonlinear, so the substitution does not transfer Theorem 1.1 to unary PHP in $\text{Res}(\oplus)$. Nor does Theorem 9.1 apply directly: no sufficiently dense separated space is known for the unary encoding.

Size and width. No relation between $\text{Res}(\oplus)$ size and width, or between size and PC degree, is claimed (Remark 9.3).

Stronger systems. A lower bound for the subsystem $\text{Res}(\oplus)$ does not by itself give a Frege lower bound. In the translation of $\text{AC}^0[p]$ -Frege proofs [BIKPRS] the extension axioms are nested: inputs of one block mention variables of others. Step 2 uses that all inputs are affine in the original variables, and we know of no removal step for nested blocks.

Other possible applications suggested in review, such as further succinctly encoded principles or resolution over linear equations modulo a prime $p > 2$, have not been examined and nothing is claimed about them.

10 Evidence and the research framework

The supplementary research record is maintained as a dated online notebook:

<https://kbr.is-a.dev/math-research/>

It presents arguments, failed attempts, and computation records from the broader project. The GitHub repository introduced above stores its source, supporting programs and data, and earlier versions. These materials document how the result was developed; the mathematical argument is presented in this manuscript.

10.1 Mathematical provenance and component checks

This manuscript presents the [formalized notebook theorem and its proof](#) [Lean], recorded on 15 September 2026, together with the [exponential bound](#) and the [generic criterion](#), formalized on 17 September 2026. Version 1 corresponds to repository revision [54f0937](#), which contains the superpolynomial theorem and its complete formal dependency route. The formal content of revision 1 is contained in [b47e9b1](#), which adds the modules for Theorem 1.1, Theorem 9.1, Corollary 9.2, and Remark 9.4. The earlier [internal publication audit](#) and subsequent literature notes in the notebook distinguish the claimed unrestricted result from the regular and depth-restricted literature.

The repository retains exact primitive-PC traces for the rule simulation, including missing-premise controls. A separate matching-moment checker stores sparse separating functionals and every potentially nonzero marginal equation for finite instances, together with falsifying perturbations. These finite experiments were development checks, not a proof of the asymptotic statement. The subsequent Lean formalization now verifies that statement and its entire dependency route, including the chessboard filling theorem, actual CNF-to-PC translation, and uniform parameter closure. The permanent [Lean] links in this revision refer to published commit [b47e9b1](#). Its project pins Lean 4.34.0-rc2 and the Mathlib dependency revision. The final transitive axiom report contains only Lean’s standard foundations; there is no `sorry`, custom axiom, or unproved chessboard interface. The compiled module of the superpolynomial theorem and all its imported declarations also passed

a fresh kernel replay with the bundled `leanchecker -fresh` on 15 September 2026; the modules added in revision 1 passed the project’s verification script, with the same axiom report, but were not separately replayed in this way. The complete project was separately rebuilt from empty project build outputs using the pinned library caches. The proofs here follow the formalization entries, including alternative arguments adopted during formalization. Formal verification of the stated objects is distinct from external assessment of the manuscript’s exposition, interpretation of the proof system, and novelty.

10.2 An inspectable workflow for autonomous research

The project also develops *Noemesis*, an open framework for autonomous mathematical research: a collection of instructions, tools, and record-keeping conventions for AI assistants, distributed in the same repository. Its purpose is to let an autonomous assistant continue a mathematical investigation across finite context windows and session boundaries while retaining inspectable evidence [Repo]. Its main mechanisms are:

- **Persistent context.** A stable restart guide, a concise living overview, and a claim index recover the relevant definitions and proofs without reloading the full conversation or depending on a particular machine.
- **Recorded successes and failures.** Dated, append-only entries retain arguments, assumptions, obstructions, and unsuccessful attempts with explicit claim status. Corrections receive new entries instead of silently replacing history.
- **Reproducible experiments.** Protected execution, complete outputs, provenance hashes, and timing records preserve what was actually computed under bounded resources.
- **Repeated investigation and process review.** Each cycle targets a named obligation, checks its outcome, and assesses wasted work or missing tools. Bounded framework improvements are recorded separately from mathematical results.
- **Explicit formal verification.** Assigned formalization cycles build dependency-indexed Lean proofs, record their exact scope and arguments, and link them from the claim index. Independent branches can run in isolated worktrees before integration; cached dependencies avoid redundant verification.
- **A live, versioned notebook.** Saved versions in Git, a version-control system, and the public notebook let readers inspect the development locally or online.

The durable record uses ordinary source, HTML, Markdown, and data files. These mechanisms make work recoverable and reviewable; they do not make a recorded proof automatically correct or formally verified.

10.3 Contributions and AI assistance

The mathematics in this paper was produced by AI models working inside the framework above, under my direction. For version 1, GPT-6 Astra (in the Codex command-line agent) was the main contributor to the mathematical development, the Lean formalization, and the drafting; Claude Sonnet 5, Claude Fable 5.1, and Claude Opus 5 were used in earlier exploratory conversations. For revision 1, Claude Fable 5.1 (in Claude Code) formalized the exponential bound of Theorem 1.1

and the generic criterion of Section 9, statements and proofs, with one batch of proofs delegated to Claude Opus 5; it also triaged reader and AI-review feedback on version 1 and drafted the revised text, including the proof overview and the related-work discussion. My own contribution is the design, direction, and iterative refinement of the research framework, the choice of problems, and steering questions about whether the work was approaching its goal; it is not the mathematical derivations. I do not have the expertise to check the mathematics myself, and I did not check it, in version 1 or in this revision. My confidence in the results rests on the Lean formalization, whose scope is the statements carrying [Lean] links, under the definitions in the linked files. The prose, the overview, the statements about the literature, and the fidelity of the formal definitions to the standard proof system are not covered by it. Neither version has been externally peer reviewed.

A first-person account of how the project developed (the chat conversations it began in, the move to file-based agents, the *Spin* research loop, and the parallel formalization) appeared as Section 8.3 of version 1. It has been moved out of the paper and is preserved in the repository as [DEVELOPMENT_HISTORY.md](#); the version-1 source containing it is fixed at commit [b47e9b1](#).

The classical topological input is attributed to [BLVZ94]; the product-extension methodology is attributed to [BIKPRS]. Original manuscript material is distributed under the [Creative Commons Attribution 4.0 International license \(CC BY 4.0\)](#).

A The required chessboard homology

This appendix proves the full homological input, including its chain-level prerequisites. All coefficients are in $\mathbb{F}_2$. The identifiers H01–H13 match the dependency route of the formalization; they are included only to make correspondence with the linked sources easy. The chessboard bound is the homological consequence of the stronger connectivity theorem of Björner, Lovász, Vrećica, and Živaljević [BLVZ94, Theorem 1.1]. We follow its star-cover strategy, with an explicit finite chain proof of the cover lemma. We do not assume that theorem or a spectral-sequence convergence result as a black box.

A.1 Augmented chains and their elementary operations

A finite augmented simplicial complex K is a downward-closed family of subsets of a finite vertex set V , containing the empty face. It need not have a vertex. Write $C_k(K)$ for coefficient functions on its faces with exactly k vertices, extended by zero to all subsets of V . Thus the cell count k is reduced homological degree $k - 1$: $C_0(K) = \mathbb{F}_2$ is the augmentation group. Write $\text{Exact}(K, k)$ when every k -cell cycle has a $(k + 1)$ -cell filler, and $\text{Acyclic}(K, n)$ when this holds for every $0 \leq k < n$. For $n \geq 1$, this says reduced homology vanishes from degree -1 through $n - 2$; for $n = 0$ it is vacuous. In particular, $\text{Exact}(K, 0)$ means that K has a vertex.

Proposition A.1 (H01: supported augmented boundary). [Lean] *For $k \geq 1$, the formula*

$$(\partial c)(T) = \sum_{v \notin T} c(T \cup \{v\})$$

defines a linear map $C_k(K) \rightarrow C_{k-1}(K)$. Its restriction from $C_1(K)$ to $C_0(K)$ is the sum of vertex coefficients. The outgoing boundary of $C_0(K)$ is zero. The supported-coefficient model is linearly equivalent to functions on exact-size faces, including for chessboard complexes.

Proof. A nonzero summand requires $T \cup \{v\} \in K$ and cardinality k . Downward closure gives $T \in K$, and $v \notin T$ gives $|T| = k - 1$. Finite sums are linear. At count zero every inserted face is nonempty,

so its coefficient vanishes. Restrict a supported function to exact-size faces and invert by extension by zero. These maps are linear and mutually inverse. $\square$

Lemma A.2 (H02: the boundary squares to zero). [\[Lean\]](#) *The augmented boundary satisfies $\partial^2 = 0$ at every cell count.*

Proof. At a face T , the double sum inserts two distinct vertices outside T . Pair the ordered insertion (u, v) with (v, u) . They contribute the same coefficient, twice, hence zero in $\mathbb{F}_2$. The outgoing augmentation boundary is already zero, so this includes all low-count cases. $\square$

Lemma A.3 (H03: injective chain maps and relabeling). [\[Lean: chain maps · relabeling\]](#) *An injective vertex map carrying faces of K to faces of L induces a degree-preserving linear pushforward commuting with boundary. If it gives a bijection on the faces of the two complexes, this is a chain isomorphism and transports cycle fillings in both directions. In particular this applies to subcomplex inclusions, chessboard transpose, and relabeling a residual board. It also identifies the supported boundary with the boundary on exact-size chessboard faces.*

Proof. On an image face $f(S)$, give the pushforward coefficient $c(S)$, and assign zero off the range. Injectivity preserves cardinality and makes the definition unambiguous. On a face outside the range all boundary terms vanish. On an image face, insertions in the image correspond bijectively to insertions in the source, and all other terms vanish. Thus pushforward commutes with boundary. A face bijection supplies the inverse coefficient map. Swapping row and column coordinates preserves precisely the matching condition; residual relabeling has the same property. Restriction to exact-size faces and extension by zero identify the two boundary formulas term by term, including augmentation. $\square$

Lemma A.4 (H04: explicit cone contraction). [\[Lean\]](#) *Suppose v is a cone vertex of K , meaning that $S \in K$ implies $S \cup \{v\} \in K$. Define*

$$(s_v c)(U) = \begin{cases} c(U \setminus \{v\}) & v \in U, \\ 0 & v \notin U. \end{cases}$$

Then $s_v : C_k(K) \rightarrow C_{k+1}(K)$ and $\partial s_v + s_v \partial = \text{id}$. Consequently $\text{Exact}(K, k)$ holds for every $k \geq 0$.

Proof. The support assertion follows by adjoining the cone vertex. If $v \notin T$, the only nonzero term in $(\partial s_v c)(T)$ is insertion of v , giving $c(T)$, and $(s_v \partial c)(T) = 0$. If $v \in T$, all insertions outside T occur once in each of $\partial s_v c$ and $s_v \partial c$ and cancel; the remaining insertion of v into $T \setminus \{v\}$ gives $c(T)$. For a cycle, the identity gives $\partial(s_v c) = c$. $\square$

Lemma A.5 (H05: the boundary of a simplex). [\[Lean\]](#) *The simplex on a nonempty finite vertex set is exact at every cell count. If that set has size $b \geq 1$, its boundary complex, consisting of all proper subsets, is exact at every k with $k + 1 < b$. Equivalently it is $\text{Acyclic}(\partial \Delta, n)$ whenever $n < b$. No top-dimensional vanishing is asserted.*

Proof. The full simplex is a cone at any vertex. Contract a k -cycle in its boundary inside the full simplex using Lemma A.4. Its filler has $k + 1 < b$ vertices on every supported face, so all those faces are proper subsets and the filler already lies in the boundary complex. $\square$

A.2 The finite homological cover argument

Let $(L_i)_{i \in I}$ be finitely many subcomplexes of K . Put $L_J = \bigcap_{i \in J} L_i$, with $L_\emptyset = K$. Its vertex-nonempty nerve is the complex

$$\mathcal{N} = \{\emptyset\} \cup \{J \subseteq I : L_J \text{ has at least one vertex}\}.$$

An empty-face-only intersection does not give a nonempty nerve face. The family covers K if every face belongs to some L_i .

Lemma A.6 (H06: cover double chains and horizontal exactness). [Lean: [cover definitions](#) · [double chains](#)] *Let $D_{a,b}$ consist of double coefficients $d(J, S)$ supported where $|J| = a$, $|S| = b$, $S \in L_J$, and $J \in \mathcal{N}$. Insertion boundaries H in J and V in S lower the respective cell counts, with zero outgoing maps at count zero, and satisfy*

$$H^2 = V^2 = 0, \quad HV = VH, \quad (H + V)^2 = 0.$$

If the L_i cover K , then for $b > 0$ every horizontally closed $d \in D_{a,b}$ has a horizontal filler in $D_{a+1,b}$.

Proof. Removing an index preserves intersection membership and nerve membership. Removing a vertex preserves membership in every complex, while the nerve condition on J remains unchanged. These facts prove the support claims. The square-zero identities are Lemma A.2; both HV and VH sum the same coefficients $d(J \cup \{i\}, S \cup \{v\})$, so interchanging finite sums proves commutation. Characteristic two gives the total square-zero identity.

For each nonempty supported face S , choose an index $i(S)$ whose member contains S , and contract the index coefficient $J \mapsto d(J, S)$ at $i(S)$, using the ambient formula of Lemma A.4. Call the result $u(J, S)$, and use zero on other S . Its nonzero coefficients have S in every indicated cover member, including the newly chosen one. Because S contains a vertex, the enlarged index set really belongs to the nerve. Thus $u \in D_{a+1,b}$, and the contraction identity gives $Hu = d$. The condition $b > 0$ is essential: the $b = 0$ edge is the nerve chain complex and is not automatically exact. $\square$

Lemma A.7 (H07: homological cover lemma). [Lean] *Suppose (L_i) covers K . Let $n \geq 0$, and assume $\text{Acyclic}(\mathcal{N}, n)$. Suppose also that for every nonempty $J \in \mathcal{N}$ and every $r \geq 0$ with $|J| + r \leq n$, $\text{Exact}(L_J, r)$ holds. Then $\text{Acyclic}(K, n)$. In reduced degrees, putting $n = q + 2$, this is the usual assertion that a q -acyclic nerve and $(q - |J| + 1)$ -acyclic intersections give a q -acyclic union, including the augmentation case $q = -1$.*

Proof. We use the finite double coefficients of Lemma A.6. First note two filling facts. On the $b = 0$ edge, $d(J, \emptyset)$ is a nerve chain; nerve exactness fills horizontally at counts $a < n$, and the filler is vertically closed. For $a > 0$, a vertically closed $d \in D_{a,b}$ with $a + b \leq n$ fills vertically in $D_{a,b+1}$: for each J of size a use the assumed filling in L_J , and use zero for the other index sets. Nonempty J ensures that all filler faces lie in K as well as the required intersection.

We claim that whenever $a + b < n$ and $Hd = Vd = 0$, there is $u \in D_{a+1,b}$ with $Hu = d$ and $Vu = 0$. Induct on b . The case $b = 0$ is the nerve-edge fact. For $b > 0$, horizontal exactness first gives $x \in D_{a+1,b}$ with $Hx = d$. Then $Vx \in D_{a+1,b-1}$ is closed in both directions. Induction gives $y \in D_{a+2,b-1}$ with $Hy = Vx$, $Vy = 0$. The vertical filling fact gives $z \in D_{a+2,b}$ with $Vz = y$, since $(a + 2) + (b - 1) = a + b + 1 \leq n$. Set $u = x + Hz$. Then

$$Hu = Hx + H^2z = d, \quad Vu = Vx + HVz = Vx + Hy = 0.$$

This proves the claim by finite induction.

Finally embed a k -cell cycle c of K , $k < n$, as $d(\emptyset, S) = c(S)$ in $D_{0,k}$, with other coefficients zero. It is closed in both directions. The claim gives $u \in D_{1,k}$ with $Hu = d$, $Vu = 0$. The singleton-intersection filling gives $z \in D_{1,k+1}$ with $Vz = u$, because $1 + k \leq n$. The chain $f(S) = (Hz)(\emptyset, S)$ belongs to $C_{k+1}(K)$, and $\partial f = (VHz)(\emptyset, \cdot) = (HVz)(\emptyset, \cdot) = c$. Thus every required cycle fills, including $k = 0$. $\square$

A.3 Chessboard stars, arithmetic, and assembly

The chessboard complex $\Delta_{a,b}$ has vertex set $[a] \times [b]$ and faces the matchings, namely sets with no repeated row or column. The closed star of a vertex v consists of faces S such that $S \cup \{v\}$ is a face. It is a subcomplex and a cone at v .

Lemma A.8 (H08: first-row star cover). [\[Lean\]](#) *If $a + 1 \leq b$, the stars of $(1, j)$, $j \in [b]$, cover $\Delta_{a+1,b}$.*

Proof. A matching using row one belongs to the star of its cell in that row. Otherwise it uses at most $a < b$ columns, leaving some column j free. It can then be extended by $(1, j)$. $\square$

Lemma A.9 (H09: intersections and residual-board chain maps). [\[Lean\]](#) *The intersection of $t \geq 2$ distinct first-row stars in $\Delta_{a+1,b}$ consists exactly of matchings avoiding row one and those t columns. It is chain-isomorphic to $\Delta_{a,b-t}$, including when $t = b$ and the latter has only its empty face.*

Proof. A matching in two different stars cannot use row one, since its row-one cell would have to be both designated cells. It cannot use a designated column in another row either, since extension by that star's cell would fail. Conversely, avoiding all those rows and columns permits every required extension. Relabel the remaining rows and columns and use Lemma A.3 to obtain the chain isomorphism. $\square$

Lemma A.10 (H10: the star nerve and singleton fillings). [\[Lean\]](#) *For $a \geq 1$, $b \geq 2$, the nerve of the first-row stars of $\Delta_{a+1,b}$ is the boundary of the simplex on its b columns. It is $\text{Acyclic}(\mathcal{N}, n)$ for every $n < b$. Every individual star is exact at every cell count.*

Proof. A proper set of columns leaves an unused column and another row; their cell belongs to every indicated star. The full set of stars has no common vertex, by Lemma A.9. Thus the nerve is exactly the proper-subset complex. Apply Lemma A.5; each singleton star is a cone, so Lemma A.4 fills its cycles. $\square$

Lemma A.11 (H11: induction parameters). [\[Lean\]](#) *For natural a, b , put*

$$\nu(a, b) = \min \left\{ a, b, \left\lfloor \frac{a + b + 1}{3} \right\rfloor \right\}.$$

It is symmetric, bounded by each side, zero if a side is zero, and at least one when both sides are positive. For $b \geq 1$, $\nu(1, b) = 1$. If $2 \leq a \leq b$, then $\nu(a, b) \leq b - 1$, and for $2 \leq t < b$,

$$\nu(a, b) + 1 \leq \nu(a - 1, b - t) + t, \quad \min\{a - 1, b - t\} < \min\{a, b\}.$$

For $s \geq 1$, $N \geq 2s - 1$ gives $\nu(s, N) = s$. Cell counts $0 \leq k < n$ correspond exactly to reduced degrees $-1 \leq k - 1 \leq n - 2$, with subtraction interpreted in the integers.

Proof. Symmetry and the basic bounds follow directly from the minimum. If $a < b$, $\nu \leq a \leq b - 1$; if $a = b \geq 2$, then $\lfloor (2b + 1)/3 \rfloor \leq b - 1$. For the recursive bound write $\nu = \nu(a, b)$. Each smaller-board entry is at least $\nu - t + 1$:

$$a - 1 \geq \nu - t + 1, \quad b - t \geq \nu - t + 1, \quad a + b - t \geq 3\nu - 1 - t \geq 3(\nu - t + 1).$$

The last inequality uses $t \geq 2$. Taking the smaller minimum proves the bound; if the right side is negative the comparison is automatic. The induction measure decreases because $a - 1 < a = \min\{a, b\}$. Finally $N \geq 2s - 1$ implies $N \geq s$ and $\lfloor (s + N + 1)/3 \rfloor \geq s$, proving the narrow-board identity. The degree translation is the integer inequality obtained by subtracting one. $\square$

Theorem A.12 (H12: homological chessboard bound [BLVZ94]). [Lean] *For $a, b \geq 1$, every k -cell cycle of $\Delta_{a,b}$ has a $(k + 1)$ -cell filler whenever $0 \leq k < \nu(a, b)$. Equivalently, $\tilde{H}_j(\Delta_{a,b}; \mathbb{F}_2) = 0$ for $-1 \leq j \leq \nu(a, b) - 2$.*

Proof. Induct strongly on $\min\{a, b\}$, transposing by Lemma A.3 so that $a \leq b$. For $a = 1$, $\nu = 1$, and a vertex fills the augmentation generator. For $a \geq 2$, cover by the first-row stars. Their nerve is acyclic through count ν , since $\nu \leq b - 1$. Consider a nonempty nerve face J of size t and a count r with $t + r \leq \nu$. For $t = 1$ the star is a cone. Otherwise $2 \leq t < b$, and its intersection is $\Delta_{a-1, b-t}$, with both sides positive. Write $\mu = \nu(a - 1, b - t)$. Lemma A.11 gives $\nu + 1 \leq \mu + t$, hence $r < \mu$. The smaller induction measure and the chain isomorphism of Lemma A.9 supply the required intersection filling. Lemma A.7 now fills every $k < \nu$ cycle of the original board. $\square$

Corollary A.13 (H13: the matching-moment filling interface). [Lean] *Let $s \geq 2$, $N \geq 2s - 1$. Every cycle on exact-size $(s - 1)$ -faces of $\Delta_{s,N}$ is the boundary of a chain on exact-size s -faces. In reduced homology,*

$$\tilde{H}_{s-2}(\Delta_{s,N}; \mathbb{F}_2) = 0.$$

For $s = 2$, the cycle hypothesis includes zero augmentation.

Proof. Lemma A.11 gives $\nu(s, N) = s$. Apply Theorem A.12 at cell count $s - 1 < s$, then use the exact-face boundary equivalence of Lemma A.3. This is precisely the boundary equation used in Theorem 4.2. $\square$

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